

No. of Functional Features Included in the Solution

Date	03 October 2026
Team ID	4
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Maximum Marks	5 Marks

Functional Features Overview:

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Personalized plan generation	7-day plan from age, weight, goal, intensity and experience; validated JSON	gemini_generator.py, routes.py	Done	Core
2	Day-card plan display	Warm-up, exercises (sets × reps, rest, form cue), cooldown; rest days highlighted; print	result.html, _plan.html	Done	Core
3	Nutrition / recovery tip	2–3 sentence goal-based tip; built-in fallback tip	gemini_flash_generator.py	Done	Core
4	Feedback-based revision	Revises the latest version from plain-language feedback (≤500 chars, delimited)	updated_plan.py	Done	Core
5	Version history	Every revision stored; view any earlier version	database.py, result.html	Done	Core
6	Admin dashboard	Password login; all users; original vs latest plan; delete user	all_users.html, admin_login.html	Done	Core
7	JSON API + Swagger	Generate, tip, plan, update, history and health endpoints at /docs	routes.py, schemas.py	Done	Additional
8	Reliability features	Retry, fallback model, demo mode, friendly error page, nothing saved on failure	gemini_client.py, demo_data.py, error.html	Done	Additional

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8 (245 automated tests + manual real-Gemini run)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Profile form with validation and loading state; day cards; version history and print
2	Backend / Logic	3	Plan generation, feedback revision, tip generation
3	Database / Storage	2	Versioned plans; cascade delete
4	API / Integration	2	Gemini structured output with fallback; JSON API with Swagger
5	Security / Authentication	2	Admin password + signed cookie; input validation and prompt-injection guard

Some features span more than one category, so category counts add up to more than 8.